

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. (EC) Examination

Nov/Dec 2012

EC304 Power Electronics

Date: 29.11.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

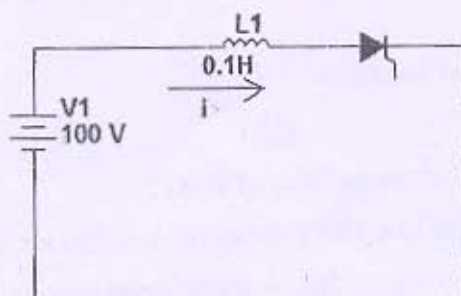
SECTION - I

Q - 1 (a) What are the problems associated with series connection of SCRs? Explain in detail. [04]
How are they eliminated?

(b) What are the different types of power diodes? Explain any one. [03]

Q - 2 (a) Justify the statement "In three phase half wave controlled rectifier, the required minimum firing angle is $\alpha=30^\circ$ ". [04]

(b) If the latching current in the circuit shown is 4mA, obtain the minimum width of the gating pulse required to properly turn-on the SCR. [04]



(c) Explain the center-tapped configuration of the single phase to single phase cycloconverter. [06]

OR

Q - 2 (a) Explain in detail how the dv/dt suppression is achieved in thyristor circuits. [04]

(b) List the various methods to turn off the thyristor. Explain any one in detail. [04]

(c) Explain the two types of control used for power transfer in ac voltage controller. [06]

Q - 3 (a) Explain the reverse recovery characteristics of diode. [04]

(b) A 100A SCR is to be connected parallel with a 150A SCR. The on-state voltage drops of the SCRs are 2.1V and 1.75V respectively. Calculate the series resistance that should be connected with each SCR if the two SCRs have to share the total current of [04]

250A in proportion to their rating. Assume that the dynamic resistance of both SCRs is equal.

- (c) Draw the circuit and output waveforms for single phase full wave converter (Mid-Point connection) with resistive load. Write the equations for i) Average dc output voltage. ii) Average load current and iii) RMS load voltage. [06]

OR

- Q - 3 (a) Draw the block diagram for closed loop control with inner current loop and field weakening. [03]
- (b) Explain the Single-pulse-width modulation method of controlling gain and output voltage of single phase inverter. [03]
- (c) Explain single phase half-wave converter for inductive load with the help of circuit diagram. Draw supply voltage, output voltage. Derive equation for average output voltage and RMS value of output voltage [08]

SECTION - II

- Q - 4 (a) Explain the ways of floating the control of gate signal with respect to ground. [03]
- (b) Explain full bridge inverter with waveforms. [04]
- Q - 5 (a) Explain the operation of 3-phase bridge inverter operating in 120° mode. Draw appropriate waveforms. [08]
- (b) Write short note 1-phase dual converter. [06]

OR

- Q - 5 (a) Explain the construction and characteristics of IGBT. [08]
- (b) A chopper circuit is connected to a 100 V dc source supplies an inductive load having 40mH in series with a resistance of 5Ω . A freewheeling diode is placed across the load. The load current varies between the limits of 10A and 12A. Determine the time ratio of the chopper. [06]
- Q - 6 (a) Explain the operating modes of DC drives. [08]
- (b) Explain the principle of step down chopper. [06]

OR

- Q - 6 (a) Draw the circuit of buckboost regulator and explain its working with equivalent circuit and waveforms. Obtain equation for ΔI . [08]
- (b) Draw and explain the block diagram for micro computer control of dc drives. [06]
